Topology-Invariant Error Dynamics and Detection-Free Cable-Fault Tolerance in Decentralized Multi-Quadrotor Cooperative Transport

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***Abstract*—When a cable snaps during multi-quadrotor cooperative payload transport, the payload drops, the freed agent accelerates away, and every surviving agent faces an instantaneous load-share jump—a 50% increase for $N{=}3$. This paper studies whether a decentralized, communication-free GPAC architecture can maintain transport without fault detection, reconfiguration, or inter-agent communication. In full-Drake simulations, the tested PID-plus-gravity-feedforward path maintains bounded post-fault tracking after one to three abrupt cable snaps, with post-fault RMSE of 43.6–48.8 cm and persistent degradation set mainly by anti-windup-limited integral action.**

The key structural observation is local and partial: in a simplified per-agent model, each agent's PID-plus-gravity-feedforward law uses only local measurements with fixed gains, so the closed-loop error dynamics share the same system matrices regardless of cable topology (Proposition 1). Cable snaps change the disturbance input but not this reduced system. This structure yields a closed-form fault-tolerance cost ratio $\rho_{\mathrm{FT}} = N/(N{-}k)$ and a finite-time transient predictor that correctly captures topology-invariant scaling, while underpredicting full-plant peaks by an approximately topology-invariant empirical factor. Ablation identifies PID feedback with gravity feedforward—not cable-tension feedforward ($<1\%$ RMSE contribution)—as the primary fault-tolerance mechanism.

Matched single-fault comparisons across $N \in \{3,5,7\}$ show post-fault RMSE decreasing from 48.8 to 38.4 cm when the fault event is held fixed. In the canonical scenarios with up to three cascading cable snaps, the same architecture maintains transport with 43.6–48.8 cm RMSE. A centralized oracle that only replaces local cable-tension feedforward with perfect post-fault load-share feedforward changes RMSE by 0.7% in the tested $N{=}3$ scenario, indicating limited benefit from that specific information path. Sensor-realistic runs (ESKF + Dryden wind), Monte Carlo campaigns (30 runs, $<3\%$ RMSE variation), and thrust-saturation sweeps support the robustness of the observed mechanism.

The topology-invariant error structure is proven only for the simplified per-agent model. Extension to the full nonlinear bead-chain plant remains a simulation-supported, partially analyzed claim; full stability and small-gain verification are open.

***Index Terms*—Cooperative aerial transport, cable-fault tolerance, N-agnostic control, PID fault tolerance, topology invariance, multi-quadrotor systems**

I. Introduction

In multi-quadrotor cooperative cable-suspended payload transport, N underactuated vehicles are connected to a shared rigid payload by flexible cables, forming a system on $\mathcal{Q} = \mathrm{SE}(3) \times (\mathrm{SE}(3) \times \mathbb{S}^2)^N$ with $6 + 8N$ degrees of freedom. Geometric control methods [1]–[4] provide almost-global exponential stability on $\mathrm{SO}(3)$ and aggressive trajectory tracking, bringing multi-vehicle cable transport close to deployment readiness [5], [6]. Yet each cable remains a single mechanical point of failure. When a cable snaps, the payload accelerates downward ($\approx$3.3 m/s^2), the freed agent accelerates upward ($\approx$6.5 m/s^2), and each remaining agent experiences an instantaneous load-share jump—a 50% increase for $N{=}3$. These effects propagate from sub-10 ms (rigid-body force transmission) to $\sim$200 ms (cable wave propagation), producing a multi-channel fault signature that surviving agents must absorb without centralized coordination or inter-agent communication.

This challenge is sharpened by the architecture's defining property. The baseline GPAC (Geometric Position and Attitude Control) hierarchy enforces an *N-agnostic constraint*: each agent knows only its own mass and inertia, local sensor measurements, cable tension and direction, and a shared reference trajectory. It does not know the team size, payload mass, or any other agent's state. This removes single points of failure and communication dependency, but also rules out model-based fault detection such as multiple-model adaptive estimation, which requires $N{+}1$ hypothesis filters. The tension-channel signal-to-noise ratio for a remote cable break is only about 1.3, indistinguishable from wind perturbations. Cable-fault tolerance in this setting must therefore work without detection, isolation, or recovery.

Despite mature geometric transport theory and extensive UAV fault-tolerance work on actuator and sensor failures [7]–[9], no published work addresses cable-fault tolerance in cooperative cable-suspended aerial transport. The closest result, Liang et al. [10], assumes known failure identity and centralized reconfiguration, both incompatible with N-agnostic architectures. Zeng et al. [11] and Gao et al. [12] observed implicit resilience to agent loss in learning-based transport, but did not formalize the mechanism or address cable-suspended underactuated systems.

The research objective emerged from an empirical observation. The predecessor system Tether_Lift [13] established that the GPAC hierarchy can coordinate cooperative lift using only local measurements under the N-agnostic constraint. The present paper (Tether_Grace) asks whether the same architectural property also yields resilience to abrupt cable-topology change.

The answer lies in PID feedback with gravity feedforward. Under cable-snap faults with no explicit fault handling, the GPAC baseline maintained bounded post-fault tracking with payload RMSE of 44–49 cm across $N \in \{3,5,7\}$. Matched single-fault experiments show that the decrease in RMSE with team size is not an artifact of increasing fault count. The mechanism is the N-agnostic PID structure: each agent

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uses gravity feedforward from its own mass, PID feedback from its position error, and anti-swing damping from local cable angular velocity. None of these terms depends on N, m_L, or any other agent's state. When a cable snaps, the surviving agents' tracking errors change through the physics, and the controller remains well-defined without adaptation or re-convergence delay. Ablation confirms that disabling cable-tension feedforward changes RMSE by less than 1% (Section VI-C), while the anti-windup analysis later shows that the remaining post-fault error is dominated by a persistent offset rather than full recovery to the nominal regime.

This shifts the research problem from *how to detect and recover from cable faults* to *whether the emergent fault tolerance can be formalized and quantified*: under what conditions does an N-agnostic controller with local cable-tension feedforward produce bounded tracking degradation across cable topologies? Figure 1 summarizes this narrowing from the broader Tether_Lift baseline to the present Tether_Grace fault-tolerance question and its current evidence boundary.

This paper addresses that question by isolating the structural property that keeps the tested controller well-defined across topology changes, deriving conservative trend tools, and validating them with comprehensive simulation evidence. The replayed full-Drake path uses the GPAC hierarchy with PID-plus-feedforward control. A post-fault geometry redistribution path is also described. Adaptive extensions ($\mathcal{L}_1$, concurrent learning) are implemented in the codebase but remain future integration work.

Two structural properties make the PID-plus-gravity-feedforward design attractive. First, PID feedback with gravity feedforward provides the core fault-absorption capability, as confirmed by ablation (Section VI-C). Second, all controller terms depend only on the agent's own state, so no agent's control law is invalidated by a topology change it cannot observe.

These properties motivate a *topology-invariance design hypothesis*: if the post-fault disturbance remains within a fault-inclusive bound and the local thrust margin is adequate, then the tracking bound is governed by controller bandwidth and disturbance size rather than cable identity. Full-Drake multibody results support this: under matched single-fault conditions (identical fault, varying N), payload RMSE decreases from 48.8 cm at $N{=}3$ to 38.4 cm at $N{=}7$. A thrust-saturation boundary sweep identifies a sharp phase transition at $f_{\max} \approx 45$ N that coincides with the theoretical post-fault hover minimum, with the default 150 N providing $3\times$ headroom.

The contribution of this paper is threefold:

C1 *Topology-invariant error structure and mechanism identification:* A structural analysis (Proposition 1) showing that the N-agnostic PID-plus-feedforward control law produces per-agent error dynamics that are identical across cable topologies in a simplified per-agent model: cable snaps change the disturbance input, not the reduced system matrices. This provides a partial structural explanation for the simulation results. A finite-time transient prediction ($\sim$18 cm predicted, observed $\sim$107 cm) is used as a topology-invariant qualitative scaling tool rather than a standalone predictor across plant configurations, and a closed-form cost ratio $\rho_{\mathrm{FT}} = N/(N{-}k)$ provides a conservative screening metric. Ablation and anti-windup analysis identify PID feedback with gravity feedforward as the primary mechanism and persistent offset as the dominant post-fault limitation.

C2 *Comprehensive multibody evidence:* Full-Drake simulation across $N \in \{3, 5, 7\}$ with 1–3 cascading cable-snap faults under ideal and sensor-realistic (ESKF + Dryden wind) conditions. Monte Carlo validation (30 runs per scenario) with randomized fault timing, wind, cable lengths, and initial conditions. Matched single-fault experiments ($48.8 \rightarrow 38.4$ cm), fault-count isolation, trajectory generalization, and baseline comparison provide deconfounding evidence.

C3 *Practical design-screening tools for the tested architecture:* A fault-tolerance cost ratio ρ_{FT} compared against empirical nominal-vs-fault RMSE ratios, a thrust-margin screening methodology validated by the $f_{\max}$ phase transition at the theoretical minimum, and a tilt-corrected capacity analysis for actuator sizing in the tested PID-plus-gravity-feedforward architecture.

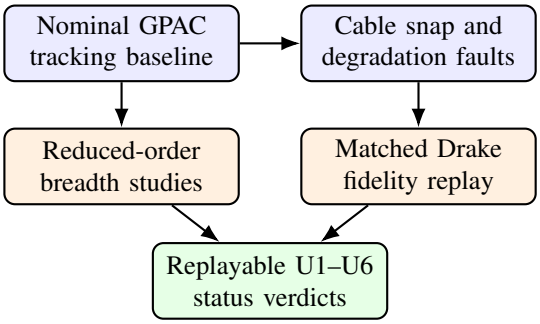


Fig. 1: Overview of the paper scope. The study begins from the validated GPAC baseline (Tether_Lift), introduces the cable-snap fault problem, documents the Tether_Grace fault-tolerance extension, and evaluates the present evidence.

The rest of the paper is organized as follows: Section II reviews related work. Section III defines the system model, the cable-snap fault event, and the formal problem. Section IV presents the fault-tolerant architecture and its integration boundaries. Section V defines the evidence methodology. Section VI reports simulation results. Section VII interprets findings, identifies gaps, and concludes.

II. Related Works

This section reviews the three areas that intersect in the paper's contribution: cooperative cable-suspended aerial transport (Section II-A), fault-tolerant control for UAVs (Section II-B), and adaptive control relevant to topology-change resilience (Section II-C). It synthesizes the literature to identify the gap: no prior work addresses cable-fault tolerance in decentralized cooperative aerial transport or formalizes the mechanism by which N-agnostic control provides implicit fault tolerance under topology changes.

A. Cooperative Cable-Suspended Aerial Transport

The geometric control foundation begins with Lee, Leok, and McClamroch [1], who developed singularity-free tracking on $\mathrm{SO}(3)$. Sreenath, Lee, and Kumar [2] extended this to cooperative cable-suspended payload manipulation on $\mathrm{SE}(3)\times(\mathbb{S}^2)^N$, Lee [3] proved almost-global exponential stability on $\mathrm{SE}(3)\times\mathbb{S}^2$ for cable-suspended rigid-body transport, and Dai, Lee, and Bernstein [4] introduced adaptive load-mass estimation.

Recent advances include agile cooperative aerial manipulation using INDI-based control and centralized NMPC [6], efficient QP-based cable-force allocation for embedded microcontrollers [14], the validated flexible simulator RotorTM [15], and a recent survey of suspended-load transport [5].

Every published controller assumes intact cables. No result addresses tracking guarantees under cable loss, provides design criteria for sizing against cable-loss disturbances, or characterizes fault tolerance in this system class.

B. Fault-Tolerant Control and Fault Detection for UAVs

The UAV fault-tolerant control (FTC) literature is extensive for actuator and sensor faults. Representative results include INDI-based flight after complete loss of two opposing rotors [7], fault-tolerant neural CBFs for sensor faults [8], and recent surveys of UAV fault detection and diagnosis [9].

Critical observation: All studied fault modes are actuator faults (rotor degradation/loss) or sensor faults (IMU bias, GPS dropout). Cable faults, which change the mechanical coupling between agents and payload, are absent.

The only paper addressing cable-related failure in multi-UAV transport is Liang et al. [10], who propose dynamic load reassignment with sliding-mode observers. However, this work assumes known failure identity, requires centralized reconfiguration, provides no detection mechanism, and includes no experimental validation, all of which are incompatible with the N-agnostic setting.

Related cable-failure work outside cooperative multi-UAV transport includes cable-driven parallel robots [16]. None addresses cooperative multi-UAV transport or satisfies the N-agnostic constraint.

C. Adaptive Control and Disturbance Estimation

Two adaptive paradigms are most relevant to topology-change resilience. Concurrent learning (CL) [17], [18] replays stored regressor-measurement pairs to achieve parameter convergence without persistent excitation, but after abrupt parameter changes its buffer becomes a liability because pre-change data bias the estimate during turnover. $\mathcal{L}_1$ adaptive control [19], [20] provides uniform transient guarantees with memoryless piecewise-constant adaptation, making it structurally suited to topology-change scenarios; Wu et al. [21] demonstrated $\mathcal{L}_1$ augmentation of geometric quadrotor control at >200 Hz. Related decentralized adaptive transport work includes fuzzy-logic adaptation [22] and adaptation to robot departure [12].

Neither CL nor $\mathcal{L}_1$ has been applied to cooperative cable transport, nor has either been analyzed under abrupt topology changes where the system structure, not just parameter values, changes instantaneously.

Control barrier functions (CBFs) [23] provide forward invariance via QP-based control modification; Input-to-State Safety (ISSf) [24] extends these guarantees to bounded disturbances, and high-order extensions [25] handle relative-degree constraints. Yang et al. [26] combined CBFs with disturbance estimators for cable-suspended payload safety. Complementary tools include contraction-theory tracking-error bounds [27]. None of these methods has been applied to post-fault topology changes in multi-vehicle cable transport.

The theoretical foundations for analyzing topology changes include stability of switched systems under average dwell-time [28], ISS under switching [29], and the hybrid dynamical systems framework [30] for continuous flow interrupted by discrete jumps—the natural setting for cable-snap events.

The closest empirical observations of emergent fault tolerance are Zeng et al. [11] and Gao et al. [12]. Zeng et al. demonstrated implicit resilience to MAV loss in multi-agent RL for cable-suspended transport with zero-shot sim-to-real transfer, but did not formalize the mechanism. Gao et al. showed adaptation to robot departure, but in fully actuated hexarotors undergoing commanded departure rather than spontaneous cable faults.

No work formalizes the mechanism by which N-agnostic control provides implicit cable-fault tolerance. The switched systems literature provides relevant tools but has not been applied to N-agnostic cable-suspended transport with instantaneous, undetected topology changes.

The contribution lies at the intersection of three well-populated but previously disconnected areas:

(i) *Cooperative cable-suspended transport* [1]–[3], [5], [6]: mature geometric control with no cable-fault handling.
(ii) *UAV fault-tolerant control* [7]–[9]: extensive actuator/sensor fault work, but cable faults are absent.
(iii) *Adaptive decentralized control* [12], [17], [19], [21]: rich theory not analyzed for topology-change resilience in cable transport.

No prior work provides tracking bounds or design criteria for cable-fault tolerance in decentralized cooperative aerial transport. The present paper addresses this gap.

III. Problem Statement

This section defines the physical system, the baseline control architecture, the cable-snap fault event, and the formal problem.

The system comprises N quadrotors cooperatively transporting a rigid payload via flexible cables. The state evolves on

$$\mathcal{Q} = \mathrm{SE}(3)\times\left(\mathrm{SE}(3)\times\mathbb{S}^2\right)^N, \quad \dim(\mathcal{Q}) = 6+8N, \quad (1)$$

where $\mathrm{SE}(3)$ captures the payload's six-DOF pose, each quadrotor contributes a further $\mathrm{SE}(3)$, and each cable direction lives on $\mathbb{S}^2$. The default configuration uses $N=3$ ($\dim(\mathcal{Q})=30$), with scaling experiments extending to $N\in\{5,7\}$.

Each quadrotor $i \in \{1, \ldots, N\}$ has mass $m_Q = 1.5$ kg and obeys (with $\boldsymbol{e}_3 = [0, 0, 1]^\top$ pointing upward and $g = 9.81$ m/s^2):

$$m_Q \ddot{\boldsymbol{p}}_i = -m_Q g\, \boldsymbol{e}_3 + f_i R_i \boldsymbol{e}_3 + \boldsymbol{F}_i^{\text{cable}} + \boldsymbol{F}_i^{\text{wind}}, \quad (2)$$

$$J\dot{\boldsymbol{\Omega}}_i = -\boldsymbol{\Omega}_i \times (J\boldsymbol{\Omega}_i) + \boldsymbol{\tau}_i + \boldsymbol{\tau}_i^{\text{ext}}, \quad (3)$$

$$\dot{R}_i = R_i \widehat{\boldsymbol{\Omega}_i}, \quad (4)$$

where $f_i \in [0, f_{\max}]$ is the scalar thrust ($f_{\max} = 150$ N), $R_i \in \text{SO}(3)$ is the body-to-world rotation, $J \in \mathbb{R}^{3\times 3}$ is the inertia tensor (from a $0.30 \times 0.30 \times 0.10$ m solid box), $\boldsymbol{\Omega}_i \in \mathbb{R}^3$ is the body angular velocity, $\boldsymbol{\tau}_i \in \mathbb{R}^3$ is the commanded torque ($|\tau_k| \leq 10$ N·m per axis), $\boldsymbol{F}_i^{\text{cable}}$ is the cable force, and $\boldsymbol{F}_i^{\text{wind}}$ is the aerodynamic disturbance.

The payload is a rigid sphere of mass $m_L = 3.0$ kg and radius $r_L = 0.15$ m:

$$m_L \ddot{\boldsymbol{p}}_L = -m_L g\, \boldsymbol{e}_3 + \textstyle\sum_{i=1}^N T_i \boldsymbol{q}_i + \boldsymbol{F}^{\text{contact}} + \boldsymbol{F}_L^{\text{wind}}, \quad (5)$$

where $T_i \geq 0$ is the scalar cable tension, $\boldsymbol{q}_i \in \mathbb{S}^2$ is the unit direction from payload to quadrotor i, and $\boldsymbol{F}^{\text{contact}}$ accounts for ground interaction (Coulomb friction, $\mu_s = 0.5$, $\mu_k = 0.3$). Cable attachment points are at radius 0.105 m on the payload surface. The payload mass m_L is *unknown to every agent*. The force balance in (5) is the plant-level relation that later explains how cable-load redistribution appears immediately at the payload during a snap event.

Each cable is modeled as a chain of $n_b = 8$ point-mass beads ($m_{\text{rope}} = 0.2$ kg) creating $n_b + 1 = 9$ spring-damper segments with tension

$$\mathcal{T}_j = \begin{cases} 0, & \text{if } \epsilon_j \leq 0 \text{ (slack)}, \\ k_s \epsilon_j + c_s \max(0, \dot{\epsilon}_j), & \text{if } \epsilon_j > 0 \text{ (taut)}, \end{cases} \quad (6)$$

where $\epsilon_j = \|\boldsymbol{b}_{j-1} - \boldsymbol{b}_j\| - L_0$ is the segment stretch and $L_0 = L_{\text{rest}}/(n_b + 1)$. Stiffness is derived from maximum allowable strain $\varepsilon_{\max} = 0.15$:

$$k_s = \frac{m_L g}{N \cdot L_{\text{rest}} \cdot \varepsilon_{\max}} \cdot (n_b + 1). \quad (7)$$

The piecewise law in (6) is the key cable nonlinearity for the reported fault transients, because it captures the slack-to-taut switching that a rigid-cable model would miss. Damping is scaled to preserve cable-level damping across discretization: a reference cable damping c_{cable} is derived from a reference configuration (n_b=8, $c_{\text{ref}} = 15$ N·s/m, $k_{\text{ref}} = 300$ N/m) and per-segment damping is $c_s = c_{\text{cable}} \cdot (n_b{+}1)$, ensuring dampers-in-series equivalence. Cable lengths are deterministic for each scenario (e.g., $[1.00, 1.08, 1.16]$ m for N=3); Monte Carlo campaigns additionally randomize lengths by $\pm 5\%$ from these baselines, producing up to 19% asymmetry.

The tension-only constraint means cables cannot push. The bead-chain discretization captures lateral vibration, wave propagation ($v_{\text{wave}} \approx 7$ m/s), distributed inertia, and slack-to-taut transitions that rigid-cable assumptions miss. The plant is simulated in Drake [31] at $\Delta t_{\text{sim}} = 0.2$ ms (5 kHz).

Aerodynamic disturbance follows the MIL-F-8785C Dryden turbulence model ($\sigma_u = \sigma_v = 1.0$ m/s, $\sigma_w = 0.5$ m/s; $L_u = L_v = 200$ m, $L_w = 50$ m; mean wind $[2.0, 0, 0]$ m/s). Spatial correlation decays as $\rho(\boldsymbol{p}_i, \boldsymbol{p}_j) = \exp(-\|\boldsymbol{p}_i - \boldsymbol{p}_j\|/l_c)$ with $l_c = 10$ m.

Each agent carries an IMU (400 Hz; gyroscope noise 5×10^{-4} rad/s/$\sqrt{\text{Hz}}$; accelerometer noise 4×10^{-3} m/s^2/$\sqrt{\text{Hz}}$), GPS (10 Hz; position noise $[0.02, 0.02, 0.05]$ m), barometer (25 Hz; 0.3 m white noise, 0.2 m correlated with 5 s time constant), and a cable tension/direction sensor (200 Hz). State estimation uses a 15-state error-state Kalman filter (ESKF) with multiplicative quaternion error, fusing IMU propagation at 400 Hz with discrete GPS and barometer updates.

A. The N-Agnostic Constraint

Each agent i knows only:

- its own mass m_Q and inertia J;
- its ESKF state estimate $(\hat{\boldsymbol{p}}_i, \hat{\boldsymbol{v}}_i, \hat{R}_i)$ and bias estimates;
- its cable tension T_i and direction $\boldsymbol{q}_i$;
- a shared reference trajectory $\boldsymbol{p}_d(t)$ (pre-broadcast, evaluated locally).

It does *not* know: the number of other agents, the payload mass, any other agent's state, or whether other agents are connected. This eliminates single points of failure and communication dependency, making the per-agent policy identical regardless of team size.

The baseline GPAC architecture is a four-layer hierarchical controller per agent with separated timescales.

1) Layer 1: Position and Anti-Swing Control (50 Hz): Computes the desired force:

$$\boldsymbol{F}_{\text{des},i} = \boldsymbol{F}_{\text{fb}} + \boldsymbol{F}_{\text{ff}} + \boldsymbol{F}_{\text{cable}} + \boldsymbol{F}_{\text{swing}} + \boldsymbol{F}_{\text{eso}}, \quad (8)$$

where $\boldsymbol{F}_{\text{fb}} = -K_p \boldsymbol{e}_p - K_d \boldsymbol{e}_v$ with gain matrices K_p, K_d; $\boldsymbol{F}_{\text{ff}} = m_Q(\boldsymbol{a}_d + g\boldsymbol{e}_3)$; cable compensation $\boldsymbol{F}_{\text{cable}} = \kappa_T(t) T_i \boldsymbol{q}_i$ with ramp $\kappa_T(t) = \min(1, (t - t_{\text{taut}})/\Delta_{\text{ramp}})$, $\Delta_{\text{ramp}} = 2$ s; anti-swing $\boldsymbol{F}_{\text{swing}} = k_q \boldsymbol{e}_q - k_\omega \boldsymbol{\omega}_q$ ($k_q = 4.0$ N/rad, $k_\omega = 2.0$ N·s/rad); and ESO feedforward $\boldsymbol{F}_{\text{eso}} = m_Q \hat{\boldsymbol{d}}_i$. The fault-tolerance evaluation (Sections IV–VI) uses uniform scalar gains $k_p = 8.0$, $k_d = 8.0$, $k_i = 0.15$ (i.e., $K_p = k_p I$, $K_d = k_d I$) to enable the per-axis eigenvalue analysis in Section IV-A. Sections IV–VI analyze the reduced controller actually used in the full-Drake fault-tolerance runs: reference-acceleration and ESO feedforward are disabled, the cable-tension ramp is replaced by a constant gain, and the matrix gains are specialized to the uniform scalar gains above. The force decomposition is therefore restated in Section IV in the form used for the reported fault-tolerance evidence.

2) Layer 2: Geometric SO(3) *Attitude Control (200 Hz):* Singularity-free tracking [1]:

$$\boldsymbol{\tau}_i = -k_R \boldsymbol{e}_R - k_\Omega \boldsymbol{e}_\Omega + \boldsymbol{\Omega}_i \times (J\boldsymbol{\Omega}_i), \quad (9)$$

with $\boldsymbol{e}_R = \frac{1}{2}\,\text{vee}(R_d^\top R - R^\top R_d)$, $\boldsymbol{e}_\Omega = \boldsymbol{\Omega} - R^\top R_d \boldsymbol{\Omega}_d$, $k_R = 8.0$ N·m, $k_\Omega = 1.5$ N·m·s. Almost-global exponential convergence holds on $\{\Psi_R < 2\} \subset \text{SO}(3)$.

3) Layers 3–4 and Safety Filter (Not Active in This Study): The full GPAC baseline additionally includes a concurrent learning mass estimator [17] (Layer 3, 200 Hz), an extended state observer [32] (Layer 4, 500 Hz), and an ISSf-CBF safety filter [23], [24] (200 Hz, enforcing tension, cable-angle, tilt, and collision constraints). These are implemented in the

Tether_Lift codebase but are *disabled* in the reported full-Drake runs (Remark 3) because the evaluation targets the minimal PID-plus-gravity-feedforward path. Their integration is discussed as future work in Section VII.

B. The Cable-Snap Fault Event

At time t_f, cable j breaks instantaneously: $T_j(t) \to 0$ for $t \geq t_f$. Three coupled effects occur.

Payload acceleration:

$$\|\Delta\ddot{\boldsymbol{p}}_L\| = \frac{T_j(t_f^-)}{m_L} \approx \frac{g}{N} \approx 3.27 \text{ m/s}^2 \quad (N{=}3). \tag{10}$$

Freed-agent acceleration:

$$\|\Delta\ddot{\boldsymbol{p}}_j\| = \frac{T_j(t_f^-)}{m_Q} \approx \frac{g \cdot m_L}{N \cdot m_Q} \approx 6.54 \text{ m/s}^2. \tag{11}$$

Load-share jump: from m_L/N to $m_L/(N{-}1)$—a 50% increase for $N{=}3$.

These effects propagate across two orders of magnitude in timescale:

Path	Mechanism	Timescale
Rigid-body transmission	Force through payload	∼7.9 ms
Elastic cable response	Spring-damper dynamics	10–50 ms
ESKF innovation	Velocity change detected	∼13 ms
ESO convergence	Third-order settling	∼33 ms
Load accel. filter	Low-pass differentiation	∼110 ms
Cable wave propagation	$v_{\text{wave}} \approx 7$ m/s	∼186 ms
Adaptive divergence	Parameter estimate drift	∼300 ms

Two Fault Scenarios Under N-Agnosticism is the fundamental problem: (A) agent i's own cable breaks, and (B) another agent's cable breaks. Scenario A is trivial to detect and isolate; Scenario B is the difficult case and the paper's focus.

1) Scenario A: Own Cable Breaks: Agent i observes $T_i \to 0$ within one sample period ($\Delta t_d = 5$ ms). Detection is trivial (infinite SNR), and position deviation during detection is negligible:

$$\|\Delta\boldsymbol{p}_i(\Delta t_d)\| \leq \frac{T_i^-}{2m_Q} \cdot \Delta t_d^2 \approx 0.082 \text{ mm}. \tag{12}$$

A threshold $T_i < T_{\min}$ (held for 0.12 s debounce) triggers an escape maneuver: climbing 0.9 m and displacing 1.8 m horizontally within 1.5 s, extending to 1.8 m climb and 3.8 m displacement within 3.0 s. Direction is biased 0.18 toward the formation tangent and 0.10 toward current velocity, chosen to limit tilt to <30° during escape and ensure adequate vertical thrust margin. Section VI-A reports the resulting separation distances for all observed faults, and Fig. 6 provides representative tilt and thrust histories in the most demanding scenario. The escape policy is therefore presented as a conservative, validated maneuver rather than as an optimized trajectory.

2) Scenario B: Another Agent's Cable Breaks: Agent i does not observe T_j directly. The fault manifests only through payload-mediated perturbations. The tension-channel SNR is

$$\text{SNR}_T^{(B)} = \frac{\Delta T_i^{(\text{imm})}}{\sigma_T} \approx \frac{4.2 \text{ N}}{3.0 \text{ N}} \approx 1.3, \tag{13}$$

where $\Delta T_i^{(\text{imm})} \approx 4.2$ N denotes the immediate tension jump measured on a surviving cable in the canonical $N{=}3$ full-Drake traces, and $\sigma_T \approx 3.0$ N denotes the corresponding pre-fault fluctuation level in that channel. The estimate is therefore empirical, drawn from the same scenario family used to motivate Scenario B, rather than a closed-form sensor-noise bound. insufficient for reliable discrimination from wind gusts. Multi-channel analysis improves SNR to ∼2–6 but with 13–300 ms delays. Scenario B is the fundamentally difficult case and this paper's focus.

Under the N-agnostic constraint, conventional FDIR is *architecturally incompatible* with Scenario B: (i) MMAE requires $N{+}1$ parallel filters, violating N-agnosticism; (ii) tension-channel SNR ≈ 1.3 is indistinguishable from wind perturbations; (iii) an N-agnostic agent has no mode parameter depending on N to reconfigure. These are structural consequences of the design choice. Any compatible fault-tolerance approach must work *without* detection, isolation, or recovery.

From agent i's perspective, translational dynamics (2) can be rewritten as

$$\ddot{\boldsymbol{p}}_i = \frac{f_i}{m_Q} R_i \boldsymbol{e}_3 + g\boldsymbol{e}_3 + \boldsymbol{d}_i(t, \boldsymbol{x}), \tag{14}$$

where $\boldsymbol{d}_i = (\boldsymbol{F}_i^{\text{cable}} + \boldsymbol{F}_i^{\text{wind}})/m_Q$ is the lumped acceleration uncertainty. This term encompasses cable tension (∼6.5 m/s^2), wind (∼0.7–2.0 m/s^2), and cable vibration (∼0.3 m/s^2). The controller never decomposes $\boldsymbol{d}_i$ into named components—this lumping enables topology invariance, since a cable snap changes the composition of $\boldsymbol{d}_i$ but not its mathematical role.

Definition 1 (Nominal and Fault-Inclusive Disturbance Sets)**.** *$\mathcal{W}_{\text{nom}} = \{\boldsymbol{d} : \|\boldsymbol{d}\|_\infty \leq \bar{d}_{\text{nom}}\}$ bounds $\boldsymbol{d}_i$ under all-cables-intact operation. $\mathcal{W}_{\text{FT}} \supset \mathcal{W}_{\text{nom}}$ additionally bounds the worst-case cable-break transient. Both bounds are derived from system parameters. The dominant contribution is cable tension. Under quasi-static hover each cable carries $T_i \approx m_L g/N$, giving a cable-tension acceleration $d_{\text{cable}} = m_L g/(N m_Q)$. Adding wind drag $d_{\text{wind}} \leq C_D A_{\text{ref}} \rho_{\text{air}} v_{\text{wind,max}}^2/(2m_Q) \approx 0.7$ m/s^2 and cable vibration $d_{\text{vib}} \leq k_s \varepsilon_{\max}/(m_Q(n_b+1)) \approx 0.3$ m/s^2 yields*

$$\bar{d}_{\text{nom}} = \frac{m_L g}{N m_Q} + d_{\text{wind}} + d_{\text{vib}} \approx \frac{3.0 \times 9.81}{3 \times 1.5} + 1.0 \approx 7.5 \text{ m/s}^2. \tag{15}$$

Post-fault, the surviving $N{-}k$ agents share the load, so the cable-tension contribution increases to $m_L g/((N{-}k)m_Q)$:

$$\bar{d}_{\text{FT}} = \frac{m_L g}{(N{-}k)m_Q} + d_{\text{wind}} + d_{\text{vib}}. \tag{16}$$

For $N{=}3$, $k{=}1$: $\bar{d}_{\text{FT}} \approx 9.81 + 1.0 \approx 10.8$ m/s^2. With safety margin for elastic transients, we use the rounded values $\bar{d}_{\text{nom}} \approx 8$ m/s^2 and $\bar{d}_{\text{FT}} \approx 12$ m/s^2. In the cable-dominated regime ($d_{\text{cable}} \gg d_{\text{wind}} + d_{\text{vib}}$), the cost ratio simplifies to a closed-form expression:

$$\rho_{\text{FT}} \approx \frac{\bar{d}_{\text{FT}}}{\bar{d}_{\text{nom}}} \approx \frac{N}{N{-}k}, \tag{17}$$

which gives $\rho_{\text{FT}} = 3/2 = 1.5$ for $N{=}3$, $k{=}1$; $5/3 = 1.67$ for $N{=}5$, $k{=}2$; and $7/4 = 1.75$ for $N{=}7$, $k{=}3$. These are

*conservative upper bounds; the empirical values (*1.26–1.39*) are lower because the fault disturbance is transient rather than sustained (Section IV-A).*

The formal problem statement is as follows: Given the system on manifold (1), the N-agnostic constraint (Section III-A), the cable-snap fault model (Section III-B), the lumped uncertainty (14) with disturbance sets (Definition 1), and the architectural incompatibility of detection-based FDIR (Scenario III-B2), the paper addresses two problems.

P1. Topology-Invariant Tracking. Identify sufficient conditions under which the per-agent tracking error satisfies

$$\|\boldsymbol{e}_i(t)\|_\infty \le \gamma_1(T_s, \omega, \bar{d}_{\mathrm{FT}}) \quad \forall\, t \ge 0 \tag{18}$$

for *all cable topologies*—where γ_1 depends on controller bandwidth and disturbance size, not on cable count or identity—provided $\boldsymbol{d}_i \in \mathcal{W}_{\mathrm{FT}}$ and thrust authority is adequate.

P2. Fault-Tolerance Cost. Derive the tradeoff

$$\rho_{\mathrm{FT}} = \frac{\gamma_1(T_s, \omega, \bar{d}_{\mathrm{FT}})}{\gamma_1(T_s, \omega, \bar{d}_{\mathrm{nom}})} > 1 \tag{19}$$

between nominal tracking and implicit fault tolerance, providing a design criterion for choosing between detection-free tolerance and detection-based recovery.

P1 is addressed by a structural analysis (Proposition 1) and comprehensive full-Drake simulation evidence. **P2** is addressed by both steady-state and transient-aware analysis, validated empirically.

IV. Fault-Tolerant Architecture and Integration Boundaries

This section presents the fault-tolerant architecture addressing problems **P1**–**P2**. The reported full-Drake fault-tolerance runs use a reduced form of the baseline GPAC force decomposition in (8): reference-acceleration and ESO feedforward are disabled, the matrix gains are specialized to uniform scalar gains, and the anti-swing law is reduced to cable-angular-velocity damping. The primary fault-tolerance mechanism in this reduced controller is PID feedback with gravity feedforward, supplemented by locally measured cable-tension feedforward and anti-swing damping. A post-fault geometry redistribution path is additionally described. The analytical core of this section is the identification of a topology-invariant error structure (Proposition 1): the N-agnostic design makes the per-agent error dynamics independent of cable topology, addressing the tracking envelope (18) and the cost ratio (19).

Each agent i computes a desired force in the world frame:

$$\boldsymbol{F}_i^{\mathrm{des}} = \boldsymbol{F}_{\mathrm{grav}} + \boldsymbol{F}_{\mathrm{fb}} + \boldsymbol{F}_{\mathrm{cable}} + \boldsymbol{F}_{\mathrm{swing}}, \tag{20}$$

where

$$\boldsymbol{F}_{\mathrm{grav}} = m_Q\, g\, \boldsymbol{e}_3, \tag{21}$$

$$\boldsymbol{F}_{\mathrm{fb}} = m_Q\big(k_p \boldsymbol{e}_p + k_d \boldsymbol{e}_v + k_i \textstyle\int \boldsymbol{e}_p\, \mathrm{d}t\big), \tag{22}$$

$$\boldsymbol{F}_{\mathrm{cable}} = \kappa_T\, T_i\, \boldsymbol{q}_i, \tag{23}$$

$$\boldsymbol{F}_{\mathrm{swing}} = -k_{\mathrm{sw}}\, \boldsymbol{\omega}_{\mathrm{cable},i}^{\perp}, \tag{24}$$

with $\boldsymbol{e}_p = \boldsymbol{p}_i^{\mathrm{ref}} - \boldsymbol{p}_i$, $\boldsymbol{e}_v = \boldsymbol{v}_i^{\mathrm{ref}} - \boldsymbol{v}_i$, and gains $k_p = 8.0$, $k_d = 8.0$, $k_i = 0.15$, $\kappa_T = 0.5$, $k_{\mathrm{sw}} = 0.5$ N·s/rad.[1]

The critical property for fault tolerance is that *every term is a function of agent i's local measurements only*. When cable j snaps, each surviving agent $i \neq j$ experiences an automatic increase in trajectory error $\boldsymbol{e}_p$ (as the payload deviates) and cable tension T_i (as load redistributes). Both signals update instantaneously through the physics, requiring no adaptation or re-convergence delay. Ablation (Section VI-C) confirms that cable-tension feedforward contributes less than 1% RMSE change; the primary fault-absorption mechanism is PID feedback with gravity feedforward.

The integral state is clamped to ± 2 m·s per axis (anti-windup). During the initial cable-snap transient ($\sim$0.5 s at $\sim$1 m peak error), the integral accumulates $\int k_i \cdot e\, \mathrm{d}t \approx 0.075$ m·s, well within the clamp. However, the post-fault load-share step $\Delta d_{\mathrm{step}} \approx 3.27$ m/s^2 (N=3, k=1, from (32)) requires a steady-state integral of $\eta_{\mathrm{ss}} = \Delta d_{\mathrm{step}}/k_i \approx 21.8$ m·s to fully compensate, far exceeding the ± 2 m·s clamp. The integral therefore saturates, and the residual uncompensated acceleration $\Delta d_{\mathrm{res}} = \Delta d_{\mathrm{step}} - k_i \cdot \eta_{\max} = 3.27 - 0.30 = 2.97$ m/s^2 is absorbed by the proportional term operating at a nonzero steady-state error $e_{\mathrm{ss}} \approx \Delta d_{\mathrm{res}}/k_p \approx 0.37$ m. This persistent offset is the dominant contributor to the post-fault RMSE increase and to ρ_{FT}. Cable-tension feedforward partially compensates this offset (the measured T_i increases post-fault), but the ablation (Section VI-C) shows its contribution is $<$1%.

The desired force is decomposed into thrust magnitude and desired body-z direction:

$$f_i = \boldsymbol{F}_i^{\mathrm{des}} \cdot \boldsymbol{R}_i \boldsymbol{e}_3, \quad \boldsymbol{b}_{3,i}^{\mathrm{des}} = \frac{\boldsymbol{F}_i^{\mathrm{des}}}{\|\boldsymbol{F}_i^{\mathrm{des}}\|}. \tag{25}$$

A. Topology-Invariant Error Analysis

The central architectural insight is that the N-agnostic control law (20) produces per-agent error dynamics that are *structurally independent of cable topology*. We formalize this as a structural property, then derive an operationally useful transient prediction.

1) Topology-Independent Error Dynamics: Consider agent i's translational dynamics under the lumped formulation (14). Under the PID law (22) with gravity feedforward (21) and ideal attitude tracking, the per-axis scalar projection of $\boldsymbol{e}_p$ (denoted e in what follows) satisfies

$$\dot{\boldsymbol{z}} = A\boldsymbol{z} + B d_i(t), \tag{26}$$

where $\boldsymbol{z} = [e,\, \dot{e},\, \eta]^\top$ with $\eta = \int_0^t e\, \mathrm{d}\tau$, and

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -k_p & -k_d & -k_i \\ 1 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \tag{27}$$

A is Hurwitz for $k_p k_d > k_i > 0$ (Routh–Hurwitz; satisfied with margin $k_p k_d = 64 \gg 0.15 = k_i$).

[1]The general GPAC baseline (Section III) uses a time-varying cable-tension ramp $\kappa_T(t) = \min(1, (t - t_{\mathrm{taut}})/\Delta_{\mathrm{ramp}})$ and a two-term anti-swing law $\boldsymbol{F}_{\mathrm{swing}} = k_q \boldsymbol{e}_q - k_\omega \boldsymbol{\omega}_q$ ($k_q = 4.0$ N/rad, $k_\omega = 2.0$ N·s/rad). For the fault-tolerance evaluation, these are simplified to a constant $\kappa_T = 0.5$ and a single cable-angular-velocity damping term $-k_{\mathrm{sw}} \boldsymbol{\omega}_{\mathrm{cable},i}^{\perp}$ ($k_{\mathrm{sw}} = 0.5$ N·s/rad).

Proposition 1 (Topology-Independent Error Structure)**.** *Under the N-agnostic PID-plus-feedforward law* (20) *with ideal attitude tracking and no actuator saturation, every agent's closed-loop error dynamics* (26) *are governed by the* same (A, B) *pair regardless of which cables are intact, the team size* N*, or the payload mass* m_L*.*

A cable snap changes the input $d_i(t)$ *(through modified cable forces propagated via the payload) but not the* system (A, B)*, because every element of* A *and* B *depends only on the PID gains* (k_p, k_d, k_i)*—none of which is a function of* N*,* m_L*, cable stiffness, or cable identity.*

This follows directly from the N-agnostic design choice. The contribution is not the observation itself but its formal consequences: the common Lyapunov function (Corollary 1), the finite-time transient prediction, and the precise identification of what remains to be proven (Gap 3). These provide the analytical framework explaining the simulation-validated fault tolerance.

Corollary 1 (Common Lyapunov Function Across Topologies)**.** *Let* $P \succ 0$ *satisfy* $A^\top P + PA = -Q$ *for some* $Q \succ 0$ *(guaranteed by* A *Hurwitz). Then* $V(\boldsymbol{z}) = \boldsymbol{z}^\top P \boldsymbol{z}$ *serves as a common ISS-Lyapunov function for the error dynamics* (26) *under* every *cable topology. No switching between Lyapunov functions is required at cable-snap events, and no dwell-time constraint applies to the sequence of topology changes.*

This is immediate from Proposition 1 (A is shared across topologies) but worth stating separately: it connects directly to the ISS switched-systems framework [28], [29] and establishes that cascading faults do not require dwell-time analysis.

Remark 1 (Hybrid System Formulation)**.** *In the framework of Goebel, Sanfelice, and Teel [30], each cable-snap event defines a jump map* $g : \boldsymbol{x} \to \boldsymbol{x}^+$ *in which* T_j *is set to zero. The continuous flow in each mode is governed by the same* (A, B) *(Proposition 1), with a common Lyapunov function (Corollary 1). Stability of the hybrid system requires three conditions: (i) each flow mode is ISS—established by* A *Hurwitz; (ii) the jump map does not increase* V *beyond recoverable bounds—requires bounding the state excursion from the cable-snap impulse (the finite-time bound below provides a partial answer, but a tight bound on* $V(\boldsymbol{z}^+) - V(\boldsymbol{z}^-)$ *from the elastic transient remains open); (iii) an average dwell-time condition holds for multiple jumps—the cascading-fault schedule (minimum inter-jump interval* 2 *s) far exceeds the* $\sim$0.5 *s transient settling time, satisfying this condition with margin. Conditions (i) and (iii) are verified; condition (ii) is the remaining open verification.*

2) Finite-Time Transient Prediction: The standard steady-state ISS gain $\gamma_{\text{ss}} = \|A^{-1}B\|\bar{d} = \bar{d}/k_i$ [33] is operationally vacuous (80 m for this system). The cable-break transient is brief, so a *finite-time* bound is far more informative. Model the fault disturbance as a pulse:

$$d_i^{\text{fault}}(t) = \begin{cases} \Delta d, & t_f \le t < t_f + \tau_{\text{fault}}, \\ 0, & \text{otherwise}, \end{cases} \tag{28}$$

where $\Delta d \approx 4.4$ m/s^2 is the cable-break impulse and $\tau_{\text{fault}} < 0.5$ s is the transient duration. The impulse response of (26) gives the transient peak:

$$\|e_{\text{peak}}\| \le \underbrace{\|\int_0^{\tau_{\text{fault}}} e^{A\tau} B \, \mathrm{d}\tau\|}_{\text{finite-time gain}} \cdot \Delta d. \tag{29}$$

For the system gains (k_p=8, k_d=8, k_i=0.15), the eigenvalues of A are $\lambda \approx \{-6.83, -1.15, -0.019\}$, all real and negative. The fast mode ($|\lambda_1| = 6.83$) decays in $\sim$0.15 s; the intermediate mode ($|\lambda_2| = 1.15$) in $\sim$0.87 s; the slow mode ($|\lambda_3| = 0.019$) governs integral wind-up with time constant $\sim$52 s. The $\mathcal{H}_\infty$ norm of the scalar transfer function $G(s) = C(sI - A)^{-1}B$ is $\gamma_{\text{ISS}} \approx 0.125$, peaked at $\omega \approx 0.15$ rad/s. Numerical evaluation of the finite-time gain matrix for $\tau_{\text{fault}} = 0.5$ s gives

$$\gamma_{\text{FT}}(\Delta d, \tau_{\text{fault}}) \approx 0.042 \cdot \Delta d \approx 0.042 \times 4.4 \approx 0.18 \text{ m}, \tag{30}$$

predicting a peak position error of $\sim$18 cm from a single cable snap—5.8$\times$ below the observed $\sim$107 cm in full Drake. The ratio $\alpha = e_{\text{peak}}^{\text{obs}}/\gamma_{\text{FT}}$ is remarkably stable across *topologies*: $107/18.4 = 5.8$ (N=3, k=1), $112/18.4 = 6.1$ (N=5, k=2), $117/18.4 = 6.4$ (N=7, k=3)—varying $< 10\%$. Across the tested topologies, this consistency suggests that the omitted effects (elastic cable coupling, attitude-loop dynamics, multi-axis interaction) can be summarized by a roughly topology-invariant aggregate correction.

The prediction captures the correct *topology-scaling*: since (A, B) are topology-independent (Proposition 1), the predicted peak is invariant to cable identity, and the full-Drake results confirm this ($\sim$107–117 cm across $N \in \{3, 5, 7\}$ with 1–3 faults, α varies $< 10\%$). However, α does *not* transfer across operating points: at higher PID gains (k_p=12, k_d=12, k_i=0.3), the observed peak increases to 116 cm despite the scalar model predicting a decrease, because the higher gains excite cable dynamics the scalar model omits. With heavier payload ($m_L = 5$ kg), the observed peak *decreases* to 100 cm despite the scalar model predicting 266 cm, because the additional payload inertia filters the transient. The scalar prediction is therefore a topology-invariant qualitative tool—it correctly predicts that peak error is cable-identity-independent and scales with Δd—but it is not a standalone predictor across different plant configurations. The paper's primary design tools (ρ_{FT}, $f_{\max}^*$, and the thrust-margin screen) do not depend on α.

3) Fault-Tolerance Cost Ratio: From the topology-independent structure, ρ_{FT} depends only on the disturbance ratio. Using the closed-form expression (17) from the cable-dominated regime:

$$\rho_{\text{FT}} \approx \frac{N}{N-k}, \tag{31}$$

which gives 1.5 for N=3, k=1. This is a conservative steady-state upper bound; the empirical values (1.26–1.39, Table I) are lower because the fault disturbance is transient, not sustained—consistent with the finite-time analysis above.

4) Gap Analysis: Simplified Model vs. Full Plant: Proposition 1 establishes topology-invariance for a scalar LTI model with exogenous bounded input. Three substantive gaps separate this from the full bead-chain Drake plant; we state each explicitly.

Gap 1: Impulsive disturbance from slack-to-taut transitions. The piecewise cable law (6) produces impulsive forces when a slack segment snaps taut. During the ∼10–50 ms elastic transient, $d_i(t)$ may momentarily exceed $\bar{d}_{\mathrm{FT}}$, violating assumption (A2). The finite-time bound (29) partially accommodates this (it integrates the actual disturbance waveform, not just its peak), but a rigorous treatment requires bounding the impulse magnitude from the cable stiffness (k_s) and segment strain ($\varepsilon_{\max} = 0.15$) and verifying that the resulting state excursion remains within the region of attraction. This is not done here.

Gap 2: Attitude-loop delay during fast force transients. Assumption (A1) requires that $R_i \boldsymbol{e}_3 \approx \boldsymbol{b}_{3,i}^{\mathrm{des}}$ instantaneously. The SO(3) inner loop runs at 200 Hz (5 ms period), but the cable-snap force transmission occurs at ∼7.9 ms (rigid-body path). During this interval, the position-loop model assumes the attitude has already aligned with the new desired force direction; in reality, the attitude lags, creating an unanalyzed residual coupling. Quantifying the resulting tracking-error increment requires a cascaded stability analysis of the two-timescale system under impulsive disturbance—a nontrivial extension beyond the scope of this work.

Gap 3: State-dependent disturbance (small-gain condition). In the full plant, $d_i(t)$ is not purely exogenous: cable forces depend on agent position relative to the payload, creating a feedback loop from $\boldsymbol{z}$ to d_i through the elastic cable dynamics. The ISS argument in Proposition 1 treats d_i as an external input. Rigorously, the interconnection of the position-loop ISS system with the cable-dynamics subsystem requires a small-gain condition [33]: the product of the position-loop ISS gain ($\gamma_{\mathrm{ISS}} = \|G(s)\|_{\mathcal{H}_\infty}$) and the cable-dynamics input-output gain (γ_{cable}) must be less than unity. A scalar (per-axis) reduction is insufficient because cable forces act along specific directions ($\boldsymbol{q}_i \in \mathbb{S}^2$), the payload inertia filters force transmission, and multiple cables distribute the coupling—all of which reduce the effective interconnection gain below the conservative scalar product. For the scalar per-axis model, the position-loop $\mathcal{H}_\infty$ gain is $\gamma_{\mathrm{ISS}} = \|G(s)\|_{\mathcal{H}_\infty} \approx 0.125$ (peaked at $\omega \approx 0.15$ rad/s), so the small-gain condition requires $\gamma_{\mathrm{cable}} < 1/\gamma_{\mathrm{ISS}} \approx 8.0$. This is a generous margin: a cable-dynamics gain of 8 would require the cable-force perturbation (in m/s^2) to exceed 8× the position error (in m), which is inconsistent with the observed bounded tracking. The full multivariable verification—computing γ_{ISS} and γ_{cable} from the linearized $6{+}8N$-dimensional plant at the taut equilibrium via Drake's `Linearize()`—is feasible with the existing infrastructure and is planned as near-term future work.

Three lines of empirical evidence support keeping the small-gain condition open rather than treating it as a disproven concern: (i) the $f_{\max}$ boundary sweep shows RMSE insensitivity for $f_{\max} \geq 45$ N (3× headroom); (ii) Monte Carlo shows <3% RMSE variation across 30 runs with randomized cable parameters; (iii) stability is maintained across all $N \in \{3, 5, 7\}$ scenarios with up to three cascading faults. None constitutes formal verification of $\gamma_{\mathrm{ISS}} \cdot \gamma_{\mathrm{cable}} < 1$, so the interconnection proof remains open.

What is established and what remains open. The shared (A, B) structure, common Lyapunov function, and hybrid-system analysis provide a coherent partial framework. Full-Drake results confirm bounded, topology-insensitive tracking degradation across $N \in \{3, 5, 7\}$ with up to three cascading faults, even though formal closure of Gap 2 remains open. To delineate the evidence boundary precisely:

- Topology-independent error dynamics (A, B) for the simplified per-agent model are *proven* (Proposition 1).
- The finite-time transient prediction ($\alpha \approx 5.8$, topology-invariant over the tested topologies) and the cost ratio $\rho_{\mathrm{FT}} = N/(N{-}k)$ are *derived analytically and compared* against full-Drake results as qualitative trend tools.
- Bounded tracking under faults in the full nonlinear Drake plant is *empirically supported* across all tested scenarios.
- The small-gain condition for cable-force feedback is *partially verified*: the scalar $\gamma_{\mathrm{ISS}} = 0.125$ is computed, but the full-plant multivariable verification remains open.
- Formal stability of the full bead-chain plant under topology switching is *open*.

Designing for $\mathcal{W}_{\mathrm{FT}}$ rather than $\mathcal{W}_{\mathrm{nom}}$ widens the tracking tube. From (17), the steady-state ratio $\rho_{\mathrm{ss}} = N/(N{-}k)$ is conservative because it assumes the fault disturbance is sustained indefinitely.

Definition 2 (Fault-Tolerance Cost Ratio)**.** *The steady-state cost ratio is $\rho_{\mathrm{ss}} = N/(N{-}k)$ (from* (17)*). A tighter* transient-aware *ratio accounts for the actual fault disturbance profile.*

5) Transient-Aware ρ_{FT}: A cable snap at t_f produces two superposed disturbances: (i) a brief impulse $\Delta d_{\mathrm{imp}} \approx 4.4$ m/s^2 lasting $\tau_{\mathrm{imp}} < 0.5$ s (the cable-break transient), and (ii) a sustained step Δd_{step} (the increased load share) that the integral term must compensate. For N agents losing k cables:

$$\Delta d_{\mathrm{step}} = \frac{m_L g}{m_Q}\left(\frac{1}{N{-}k} - \frac{1}{N}\right). \tag{32}$$

The step persists until the integral term converges, with effective time constant $\tau_{\mathrm{int}} \approx 1/k_i \approx 6.7$ s for these gains.

The transient RMSE increase is computed from the error dynamics (26) driven by the combined input $d_{\mathrm{fault}}(t) = \Delta d_{\mathrm{imp}} \cdot \mathbf{1}_{[0,\tau_{\mathrm{imp}}]} + \Delta d_{\mathrm{step}}$. Since this is additive to the nominal disturbance and approximately uncorrelated (the dominant contribution is the load-share step Δd_{step}, which is approximately constant during the integral recovery period and therefore weakly correlated with the trajectory-phase-dependent nominal disturbance), the post-fault RMSE satisfies

$$\rho_{\mathrm{FT}} \approx \sqrt{1 + \frac{\mathrm{RMSE}^2_{\text{fault-response}}}{\mathrm{RMSE}^2_{\mathrm{nominal}}}}, \tag{33}$$

where $\mathrm{RMSE}_{\text{fault-response}}$ is computed by numerically integrating (26) with $d_{\mathrm{fault}}(t)$ over the post-fault horizon ($T{-}t_f = 23$ s). For this system:

	Δd_{step} [m/s^2]	Fault resp. RMSE [cm]	ρ_{FT} (predicted)	ρ_{FT} (empirical)
N=3, k=1	3.27	33.7	1.39	1.39
N=5, k=2	2.62	27.2	1.27	1.33
N=7, k=3	2.10	22.0	1.18	1.26

The transient-aware prediction matches the empirical ρ_{FT} closely for N=3 (1.39 vs. 1.39) and underpredicts for larger N because the scalar model omits multi-fault interaction effects (the second and third cable snaps occur during the recovery transient from the first). The key insight is that ρ_{FT} is governed by the *transient energy* of the fault response—primarily the sustained step until the integral converges—not the steady-state disturbance bound. The integral time constant $1/k_i \approx 6.7$ s sets the recovery timescale and thereby determines the cost ratio.

For actuator sizing, the conservative $\rho_{\text{ss}} = 1.5$ remains appropriate (it bounds the worst case). For performance prediction, the transient-aware (33) is the operationally relevant metric.

The topology-invariance property (Proposition 1) applies to the PID-plus-feedforward law without adaptive estimation. The GPAC baseline includes a Concurrent Learning (CL) mass estimator (Layer 3) and an Extended State Observer (ESO, Layer 4), both of which are *disabled* in the reported full-Drake runs. This choice is deliberate: the topology-invariant error structure explains why PID feedback with gravity feedforward already suffices (Section VI-C).

With CL active, the feedforward includes an adaptive term $\hat{m}_L \cdot g$ that depends on the regressor history. After a cable snap, pre-fault data in the M=50 regressor buffer biases $\hat{m}_L$ toward the pre-fault load share, temporarily injecting incorrect feedforward for 2–5 s of buffer turnover. The per-agent error dynamics are no longer topology-independent in the sense of Proposition 1: the system matrix acquires an implicit dependence on cable topology through the history-dependent estimate. Topology invariance holds strictly only for the PID-plus-gravity-feedforward path. Adaptive estimation (CL, ESO, or $\mathcal{L}_1$) remains a candidate residual channel for scenarios requiring tighter steady-state tracking; its integration is discussed in Section VII.

Remark 2 (Geometry Redistribution)**.** *The codebase includes a post-fault geometry redistribution module that re-centers surviving agents' formation offsets when a cable fault is detected locally. This module operates upstream at the reference level and does not modify the GPAC control law. All reported results use fixed formation offsets throughout, demonstrating that PID-plus-gravity-feedforward alone suffices for fault tolerance. Ablation of the redistribution path is deferred to future work.*

The fault-tolerant architecture is summarized in Fig. 2. The PID-plus-feedforward control law (20), whose topology-invariance is established by Proposition 1, is the sole active mechanism in the reported runs. It does not modify the baseline GPAC internals.

V. Experimental Methodology and Evidence Sources

This paper distinguishes three levels of claim: *implemented* modules that exist in the codebase, *reproducible* results reproducible from canonical artifacts, and *open* hypotheses whose formal proof is incomplete (see the evidence-level summary in Section IV). The full-Drake simulations use the GPAC hierarchy's PID-plus-feedforward control law (Section IV) with PID feedback, gravity feedforward, and cable-tension feedforward as the fault-tolerance mechanism. The ESO feedforward and adaptive estimation are disabled. The reduced-order comparisons retain the CL baseline as the standardized reference. The related code, including the full-Drake scenarios, experiment drivers, and analysis scripts used in this study, is available in the public Tether_Grace repository [34]. That repository serves as the canonical source for the reproducible artifacts referenced throughout this section.

For the controller-comparison study, the *centralized oracle* is intentionally narrow: at the fault time it replaces local cable-tension feedforward with perfect post-fault load-share feedforward on each surviving agent. It does *not* reconfigure formation geometry, reset integral states, or modify the reference trajectory. The oracle should therefore be read as a feedforward-information baseline, not as a globally re-optimized centralized controller.

A. Scenario Families and Evaluation Pipeline

The primary full-Drake scenario families are:

(a) canonical abrupt cable faults for $N \in \{3, 5, 7\}$;
(b) matched single-fault comparison across $N \in \{3, 5, 7\}$ with identical fault event (cable 0 at $t = 7.0$ s), deconfounding team size from fault count;
(c) cable-tension feedforward ablation ($\kappa_T \in \{0, 0.25, 0.5, 1.0\}$) in N=3;
(d) thrust saturation boundary sweep ($f_{\max} \in \{35, 38, 45, 60, 80, 150\}$ N) in N=3;
(e) controller comparison hierarchy including centralized oracle, reactive FTC, and gain-scheduled PID baselines;
(f) ESO-active evaluation to assess ρ_{FT} sensitivity to controller complexity;
(g) broadened Monte Carlo with randomized payload mass ($\pm 10\%$), PID gains ($\pm 10\%$), and fault timing (± 2 s);
(h) cable discretization sensitivity: bead count $n_b \in \{2, 4, 8, 12, 16\}$ in the N=5 cascading-fault scenario, holding cable-level stiffness and damping invariant to isolate the effect of spatial resolution on transient load dynamics.

Complementary reduced-order studies—CL fidelity comparison, history-management diagnostics, gradual degradation, and non-CL adaptive law search—are reported in the supplementary material.

The primary metric is post-fault payload tracking RMSE, available across all standardized scenarios. For fidelity comparison, the key statistic is the full-Drake to reduced-order RMSE ratio under matched conditions. For history corruption, the main quantities are CL bias duration and stack refill time.

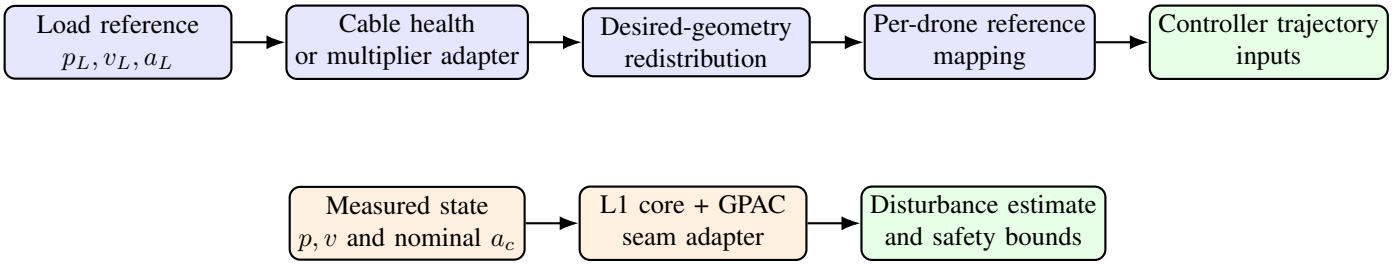


Fig. 2: Fault-tolerant signal flow. PID-plus-feedforward control (Proposition 1) with load-tracking integral on payload GPS closes the loop on actual payload position. Geometry redistribution (dashed) is implemented but not active in the reported runs.

For topology-margin screening, capacity and effective-capacity ratios are reported rather than closed-form proofs.

The evidence is organized in two tiers: (i) ideal-condition runs (plant state, no wind) establishing the core mechanism and scalability, and (ii) sensor-realistic runs (ESKF + Dryden wind) and Monte Carlo campaigns establishing robustness. Both tiers report RMSE and peak error ($\mathcal{L}_\infty$) to enable comparison with the theoretical bound (Proposition 1). The broadened Monte Carlo study is smaller (5 runs) and is interpreted accordingly as exploratory evidence rather than as the same statistical tier as the 30-run campaigns.

VI. Simulation Results

This section reports full-Drake multibody simulation results across $N \in \{3, 5, 7\}$ with abrupt cable-snap faults. All results use the Drake [31] plant at $\Delta t_{\text{sim}} = 0.2$ ms with bead-chain cable dynamics and 30 s figure-eight trajectories. Specifically, each physical cable is modeled as a chain of 8 point-mass beads connected by 9 tension-only spring-damper segments. This high-fidelity representation captures realistic elastic tension surges and slack behavior during abrupt severing faults, distinguishing it from the simplified scalar topology used for the analytical bounds.

Remark 3 (System Under Test)**.** *The full-Drake results use Layers 1–2 of the GPAC hierarchy (PID position control with gravity/cable-tension feedforward and geometric* $\mathrm{SO}(3)$ *attitude control). Layer 3 (CL mass estimator), Layer 4 (ESO), and the* $\mathcal{L}_1$ *adaptive path are* ***disabled****. The CBF safety filter is* ***disabled****. Sensor state is* ***plant-state-direct*** *(ESKF disabled). Wind disturbance is* ***disabled*** *in the ideal runs; ESKF and Dryden wind are* ***active*** *in the sensor-realistic evaluation.*

The per-agent controller uses PID position feedback ($k_p = 8.0$, $k_d = 8.0$, $k_i = 0.15$), gravity feedforward, cable-tension feedforward ($\kappa_T = 0.5$), anti-swing damping ($k_{\text{sw}} = 0.5$), and geometric $\mathrm{SO}(3)$ attitude tracking—all N-agnostic by construction. Three canonical scenarios of increasing difficulty are evaluated:

(i) *Three drones, single fault:* $N = 3$, cable lengths $[1.00, 1.08, 1.16]$ m. Cable 0 severed at $t_f = 7.0$ s. Post-fault: 2 agents, 50% load-share increase.

(ii) *Five drones, two cascading faults:* $N = 5$, cable lengths $[1.00, 1.05, 1.10, 1.14, 1.18]$ m. Cable 1 at 7.0 s, cable 3 at 12.0 s. Post-fault: 3 agents, 67% increase.

TABLE I: Full-Drake payload tracking performance: nominal and fault scenarios.

Scenario	N	Load RMSE [cm]	Peak error [cm]	$\rho_{\text{FT}}^{\text{emp}}$
Three drones, no fault	3	35.12	110.57	—
Five drones, no fault	5	34.84	110.10	—
Seven drones, no fault	7	34.70	108.65	—
Three drones, 1 fault	3	48.84	107.44	1.39
Five drones, 2 faults	5	46.37	112.30	1.33
Seven drones, 3 faults	7	43.60	117.45	1.26

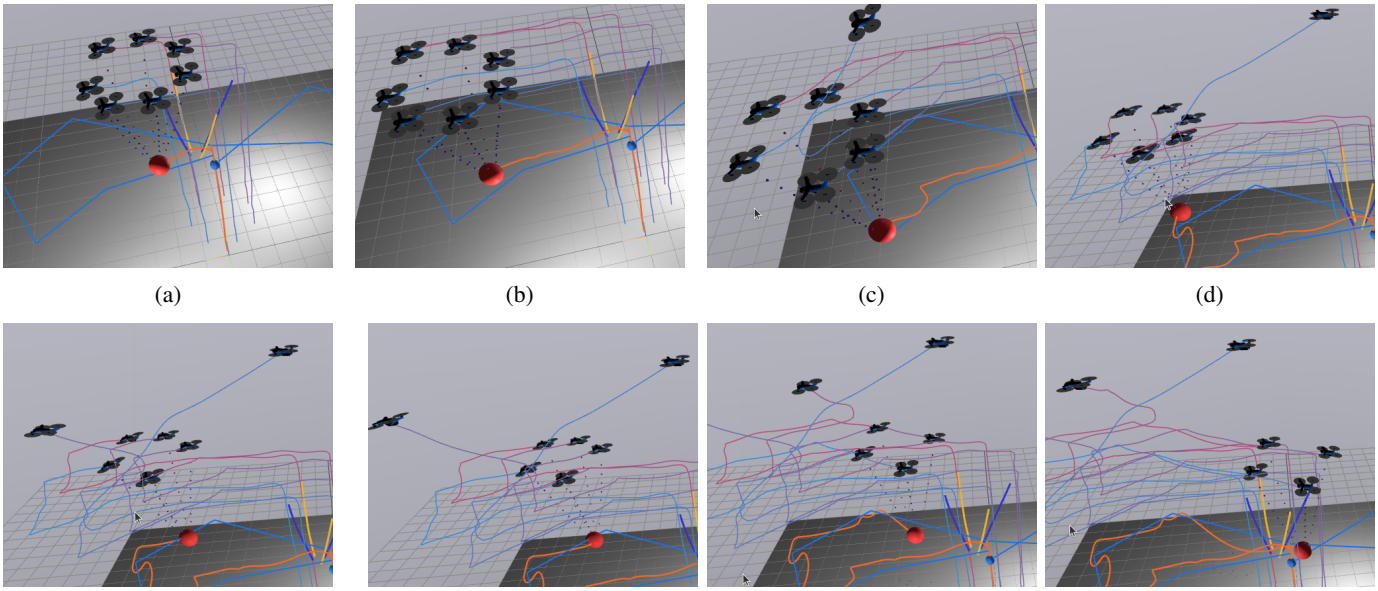
(iii) *Seven drones, three cascading faults:* $N = 7$, cable lengths $[0.98, 1.01, 1.04, 1.07, 1.10, 1.13, 1.16]$ m. Cables 1, 3, 5 at 7.0, 12.0, 14.0 s. Post-fault: 4 agents, 75% increase.

No agent is informed of topology changes except the freed agent itself, which detects its own cable loss via $T_i < T_{\min}$.

A. Canonical Fault Scenarios

Table I reports payload tracking performance across all canonical scenarios. The $\sim$35 cm nominal RMSE reflects PID-only control by design; the full GPAC stack (CL+ESO) would reduce it but would introduce history-dependent terms breaking the topology-invariant structure. The favorable ρ_{FT} values (1.26–1.39) partly reflect this choice (larger denominator); with the full stack, ρ_{FT} increases modestly (Section VI-C). The absolute post-fault RMSE (44–49 cm) is the operationally relevant quantity: it is about three payload radii and roughly 40% of a nominal 1.0–1.2 m cable length, so the controller preserves transport continuity rather than precision placement. Empirical $\rho_{\text{FT}}^{\text{emp}}$ falls below the conservative bound $N/(N-k)$ from (17), consistent with the transient-aware analysis in Section IV-A. The canonical scenarios themselves confound team size with fault count; the matched single-fault comparison below isolates the team-size effect.

The canonical scenarios suggest that average payload RMSE falls with team size even while the number of faults increases: the N=7 scenario absorbs three cascading faults yet achieves 11% lower RMSE than N=3 with one fault. Peak errors (107–117 cm) settle within 1–2 s (Figs. 4, 7), and each snap produces a bounded reconfiguration rather than loss of transport (Fig. 3). The matched single-fault comparison below shows that the same trend persists once the fault schedule is held fixed.



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 3: Seven-drone scenario ($N = 7$, cables 1, 3, 5 at 7.0, 12.0, 14.0 s). (a) at the start of the mission. (b) about 1 s before first fault. (c) about 1 s after first fault. (d) about 1 s before second fault. (e) about 1 s after second fault. (f) about 1 s before third fault. (g) about 1 s after third fault. (h) Settled after all faults: Q_0, Q_2, Q_4, Q_6 remain.

TABLE II: Fault detection and freed-agent separation.

Scen.	Drone	t_f [s]	Detect. [s]	Healthy separation [m] at fault	+2 s	final
N=3	0	7.0	0.12	0.90	1.27	2.26
N=5	1	7.0	0.12	0.66	1.07	2.69
	3	12.0	0.12	0.65	1.96	4.68
N=7	1	7.0	0.12	0.50	1.05	2.06
	3	12.0	0.12	0.50	1.90	4.73
	5	14.0	0.12	0.50	1.69	3.82

Cable tensions (Fig. 5) confirm the load-redistribution mechanism: when a cable snaps, remaining cables exhibit a tension surge as they absorb the redistributed load. The freed agent detects the fault via $T_i < T_{\min}$ within the 0.12 s debounce window and initiates escape. Detection latency is consistent across all six fault events (Table II), with all freed agents reaching 2.06–4.73 m final separation. Progressive tension oscillation growth reflects cable pendulum near-resonance ($\sim$2 s period vs. $\sim$3 s trajectory corner interval); all tensions remain positive, with peaks ($\sim$25 N) well within limits ($f_{\max} = 150$ N).

Vehicle-level dynamics for the most demanding case (N=7, Fig. 6) confirm that the GPAC hierarchy maintains stability throughout: after cascading faults, drones 1, 3, 5 escape while drones 0, 2, 4, 6 continue tracking. Formation deviation shows bounded steps at each fault, tilt stays within safe limits, and peak thrust ($\sim$40 N) remains well below $f_{\max} = 150$ N. The cross-scenario overlay (Fig. 7) confirms comparable pre-fault tracking across team sizes, localized 40–60 cm transients per fault event, and stable control even in the N=7 stress test where the third fault occurs before full recovery from the second. While run-averaged RMSE improves with N, terminal error at $t = 30$ s is 42.36, 52.79, 54.62 cm—larger teams improve average tracking but the multi-fault schedules do not produce the smallest end-of-run residual.

TABLE III: Matched single-fault comparison: identical fault event across team sizes.

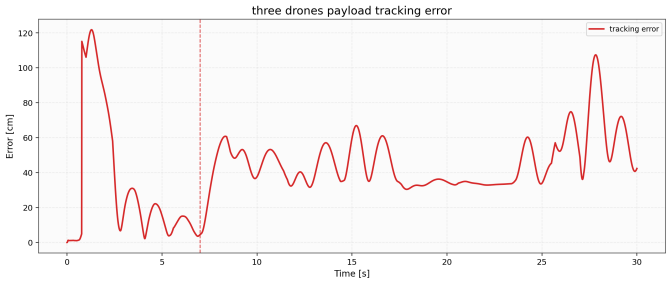
	N=3	N=5	N=7
Post-fault capacity ratio	2/3	4/5	6/7
Load RMSE [cm]	48.84	40.37	38.38
Peak error [cm]	107.44	112.09	111.15
Final error [cm]	42.36	53.25	49.85

B. Deconfounding by Team Size, Fault Count, and Thrust Margin

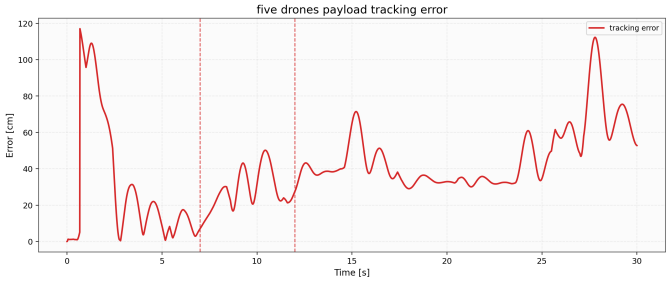
The canonical scenarios confound team size with fault count (1, 2, 3 faults for $N = 3, 5, 7$). Two complementary experiments isolate these variables.

Matched single-fault comparison. Applying the *same single-fault event* (cable 0 at $t = 7.0$ s) across all three team sizes (Table III), RMSE decreases monotonically: $48.84 \rightarrow 40.37 \rightarrow 38.38$ cm (21% improvement from N=3 to N=7), confirming that larger capacity ratios translate directly to better tracking. Peak error is nearly constant ($\sim$107–112 cm), set by the cable-snap impulse magnitude. The RMSE improvement comes from faster mid-trajectory recovery, not better final convergence (N=3 achieves the lowest terminal residual at 42.36 cm).

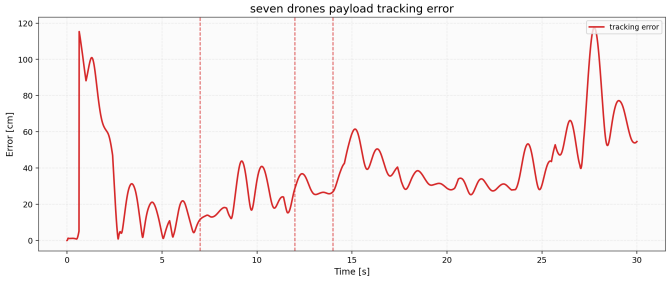
Fault-count isolation. Holding N=7 constant while varying $k \in \{1, 2, 3\}$ (Table IV), RMSE increases monotonically: 34.70, 39.39, 41.29, 43.60 cm for $k = 0, 1, 2, 3$. Each additional fault adds $\sim$2–5 cm RMSE, confirming incremental degradation rather than catastrophic failure. Peak error is



(a) $N = 3$, single fault at $t = 7.0$ s.



(b) $N = 5$, two cascading faults.



(c) $N = 7$, three cascading faults within 7 s.

Fig. 4: Payload tracking error across scenarios. Each fault produces a bounded transient settling in 1–2 s without explicit fault handling.

TABLE IV: Fault-count isolation: N=7 with 1, 2, and 3 faults.

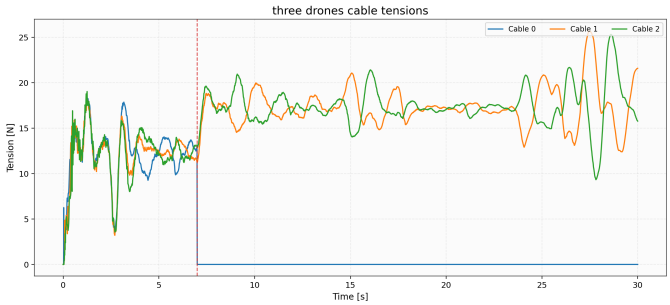
Faults k	Surviving agents	Capacity ratio	RMSE [cm]	Peak [cm]	Final [cm]
0 (nominal)	7	—	34.70	108.65	43.73
1	6	6/7	39.39	111.26	54.43
2	5	5/7	41.29	118.21	54.74
3	4	4/7	43.60	117.45	54.62

highest for k=2 (118.21 cm), likely because the second fault at t=12 s occurs during an unfavorable trajectory phase.

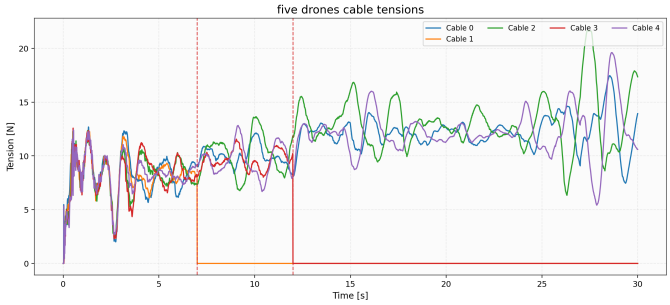
Thrust-margin verification. All scenarios satisfy the thrust-margin condition with capacity ratios 1.75–2.33 (Table V). Peak tilt during post-fault transients (15°–25°, Fig. 6c) reduces effective vertical thrust by 3–9%; all scenarios remain well above unity with this correction. A conservative design should use $f_{\max}\cos(\theta_{\max})$ as the effective vertical thrust authority.

C. Mechanism Identification

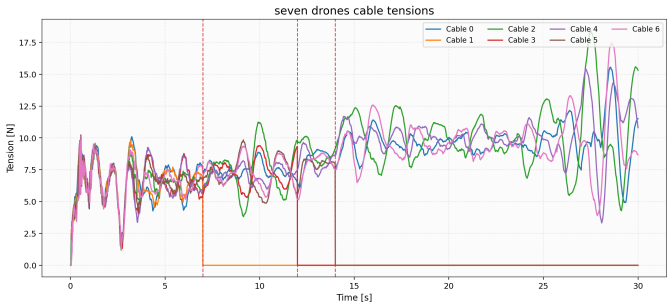
Three experiments identify what drives the fault-absorption mechanism and where it breaks.



(a) $N = 3$. Cable 0 snaps at $t = 7.0$ s; cables 1, 2 absorb the load.



(b) $N = 5$. Sequential faults at 7.0 s and 12.0 s.



(c) $N = 7$. Three sequential faults; four surviving cables remain within limits.

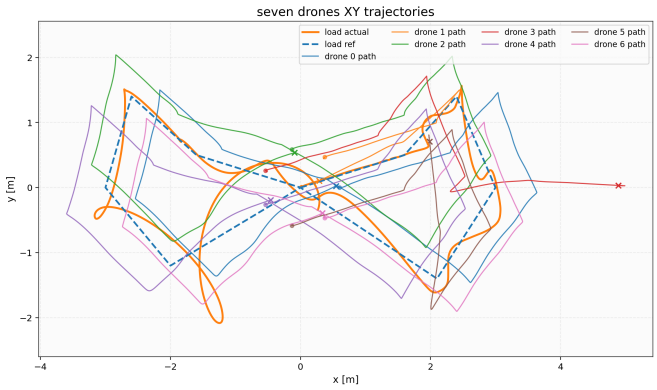
Fig. 5: Cable tension dynamics across scenarios. Remaining cables exhibit tension surges upon load redistribution; all tensions remain positive and within limits.

TABLE V: Topology-invariance thrust-margin screen.

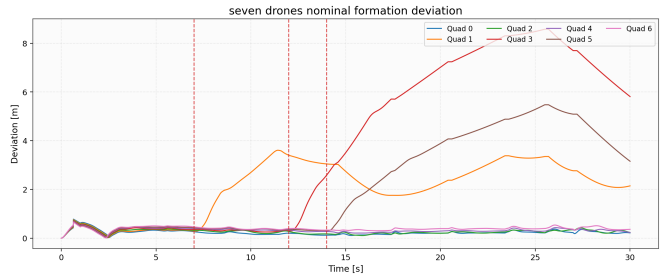
Scenario	N	Required [N/agent]	Available [N/agent]	Capacity ratio	Healthy fraction
N=3, 1 fault	3	29.43	51.50	1.75	0.67
N=5, 2 faults	5	24.53	51.50	2.10	0.60
N=7, 3 faults	7	22.07	51.50	2.33	0.57

Tension feedforward ablation. Disabling cable-tension feedforward in the most stressed scenario (N=3, single fault) changes RMSE by only 0.44 cm (0.9%) and peak error by 0.06 cm (Table VI). Varying κ_T across $\{0, 0.25, 0.5, 1.0\}$ produces total RMSE variation of 1.55 cm. This identifies the *primary* fault-absorption mechanism as PID feedback with gravity feedforward—not cable-tension feedforward. Cable-tension feedforward provides a marginal improvement (0.77 cm in final error) that is helpful but not essential.

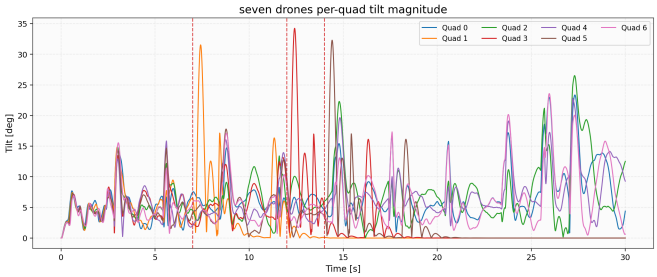
Thrust saturation boundary. Sweeping $f_{\max}$ from 150 N down to 35 N reveals a sharp phase transition (Table VII):



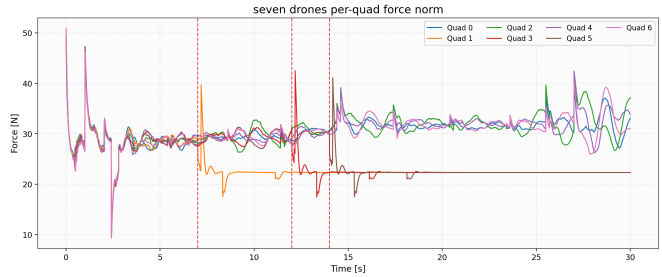
(a) XY trajectories. Drones 1, 3, 5 depart; 0, 2, 4, 6 continue.



(b) Formation deviation. Each fault produces a bounded step.



(c) Vehicle tilt. Remains within safe limits through all fault transients.



(d) Thrust per vehicle. Post-fault ∼40 N, well below $f_{\max} = 150$ N.

Fig. 6: Vehicle-level dynamics for $N = 7$ with three cascading faults. The GPAC hierarchy maintains stability: tilt stays within safe limits and thrust remains well below saturation.

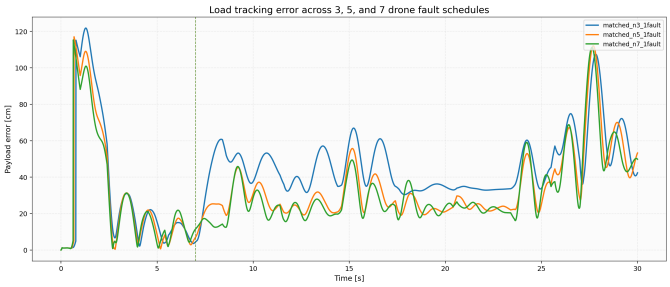


Fig. 7: Payload tracking error overlay across $N = 3, 5, 7$. Fault responses remain localized, settling on 1–2 s timescales.

TABLE VI: Tension feedforward ablation (N=3, single fault at $t = 7.0$ s).

Configuration	RMSE [cm]	Peak [cm]	Final [cm]
Full feedforward ($\kappa_T = 0.5$)	48.84	107.44	42.36
No tension feedforward	48.40	107.50	43.13

for $f_{\max} \in [45, 150]$ N, RMSE stays within 48.8–49.1 cm; at $f_{\max} = 38$ N, it jumps to 236.9 cm (4.9× degradation). This coincides with the theoretical minimum:

$$f^*_{\max} = \frac{(N \cdot m_Q + m_L) \cdot g}{N - k} + f_{\text{track}} \approx 36.8 + 8 \approx 45 \text{ N}, \tag{34}$$

where $f_{\text{track}} \approx m_Q \cdot a_{\max} \approx 8$ N accounts for figure-eight tracking demand (matching the observed pre-fault thrust above hover). The default $f_{\max} = 150$ N provides 3× headroom, confirming that the claimed fault tolerance does not depend on aggressive actuator sizing. Below the transition, RMSE is non-monotonic (236.9 cm at 38 N vs. 192.1 cm at 35 N): the system enters distinct saturation-dominated failure modes where the specific trajectory phase at the fault instant determines which agents saturate and for how long.

TABLE VII: Thrust saturation boundary (N=3, single fault at $t = 7.0$ s).

$f_{\max}$ [N]	$f_{\max}/f_{\text{hover}}$	RMSE [cm]	Peak [cm]	Final [cm]
150	10.2×	48.84	107.44	42.36
80	5.4×	48.84	107.44	42.36
60	4.1×	48.84	107.44	42.36
45	3.1×	49.06	107.66	44.10
38	2.6×	236.92	324.97	252.35
35	2.4×	192.11	250.34	179.28

Controller comparison hierarchy. To contextualize the passive GPAC approach, three alternative controllers are evaluated on the N=3 single-fault scenario (Table VIII): a centralized oracle with perfect post-fault load-share feedforward, a gain-scheduled PID triggered by elevated cable tension, and a threshold-triggered reactive controller that boosts thrust by +30%. The centralized oracle performs comparably to passive GPAC (49.18 vs. 48.84 cm, +0.7%), but this should be interpreted narrowly: the oracle only replaces the local cable-tension feedforward term and does not reconfigure geometry, reset the integral state, or alter the trajectory. The absolute difference is 0.34 cm, smaller than the 0.49 cm run-to-run

TABLE VIII: Controller comparison (N=3, single fault). The centralized oracle uses perfect post-fault load-share feedforward only; it does not alter geometry, trajectory references, or integral states.

Controller	RMSE [cm]	Peak [cm]	Detection	Communication
Centralized oracle	49.18	107.33	Perfect	Yes
Gain-scheduled PID ($1.5\times k_p, 1.25\times k_d$)	48.96	106.75	Threshold	No
Passive GPAC	48.84	107.44	No	No
Threshold-triggered	49.45	107.66	Threshold	No

TABLE IX: ESO-active evaluation: sensitivity of ρ_{FT} to controller complexity.

Scenario	RMSE [cm] Nominal	RMSE [cm] Fault	ρ_{FT} ESO	ρ_{FT} PID-only
N=3, 1 fault	28.94	40.43	1.40	1.39
N=5, 2 faults	30.50	44.59	1.46	1.33
N=7, 3 faults	33.67	46.01	1.37	1.26

standard deviation reported for the same N=3 fault scenario in Table XI. The tested detection-based alternatives therefore provide no practically meaningful benefit over the N-agnostic PID-plus-feedforward structure, but this comparison does not rule out stronger centralized policies.

ESO-active evaluation. To assess whether the favorable ρ_{FT} values are an artifact of the deliberately constrained controller, the ESO disturbance feedforward (Layer 4) is activated in all canonical scenarios (Table IX). ESO reduces nominal RMSE by 17–18%, confirming that PID-only nominal RMSE was inflated by uncompensated cable dynamics. The ρ_{FT} values increase modestly (1.37–1.46 vs. 1.26–1.39), but absolute post-fault RMSE with ESO (40–46 cm) is *lower* than PID-only (44–49 cm)—the ρ_{FT} increase reflects proportionally larger nominal improvement, not degraded fault tolerance. The topology-invariance property does not hold strictly with ESO active (history dependence); the PID-only evaluation remains the correct test.

D. Robustness Validation

Four complementary evaluations establish robustness beyond the ideal-condition canonical runs.

Sensor-realistic evaluation. Activating the ESKF and Dryden wind model produces moderate, consistent RMSE degradation: +9.9% for N=3, +8.1% for N=5, and +10.7% for N=7 (Table X). All scenarios remain stable with RMSE below 54 cm, and the uniform ∼10% degradation confirms robustness to sensor noise and wind perturbations. Peak errors increase by 12–14 cm, attributable to wind compounding fault transients and ESKF estimation lag.

Monte Carlo validation. A campaign of $N_{\mathrm{MC}} = 30$ runs per scenario randomizes cable lengths (±5%), wind seeds, and simulation seeds while using the same fixed fault schedule. Standard deviations of 0.49–1.34 cm (Table XI) confirm robustness to plant-parameter variation (< 3% RMSE variation across 30 runs). MC means are consistent with the deterministic values in Table I (within 1%), confirming representativeness. The 95th-percentile peak errors (108–120 cm) remain well within thrust-margin headroom.

TABLE X: Sensor-realistic evaluation

Scenario	RMSE [cm] Ideal	RMSE [cm] ESKF+wind	Peak error [cm] Ideal	Peak error [cm] ESKF+wind
N=3, 1 fault	48.84	53.68	107.44	119.14
N=5, 2 faults	46.37	50.14	112.30	125.17
N=7, 3 faults	43.60	48.28	117.45	131.56

TABLE XI: Monte Carlo results (mean ± std over N_{MC} runs).

Scenario	N_{MC}	RMSE [cm] mean ± std	Peak [cm] mean ± std	95th pctl. peak [cm]
N=3, 1 fault	30	48.84 ± 0.49	107.28 ± 0.47	107.85
N=5, 2 faults	30	46.40 ± 0.49	112.23 ± 1.07	113.89
N=7, 3 faults	30	43.95 ± 1.34	117.55 ± 1.37	119.84
N=3, ESKF+wind	30	53.62 ± 0.83	118.83 ± 0.91	120.19

Broadened Monte Carlo. A broadened campaign additionally randomizes payload mass (±10%), PID gains (±10%), and fault timing (±2 s). In 5 runs for N=3, broadened RMSE is 48.04 ± 1.29 cm (vs. narrow MC 48.84 ± 0.49 cm): standard deviation increases ∼2.6× but all runs remain stable. Because this broadened campaign contains only five trials, it is interpreted here as exploratory evidence of robustness to simultaneous plant-parameter variation rather than as the same statistical tier as Table XI.

Trajectory generalization. Testing across hover, point-to-point, and figure-eight references in the supplementary material confirms that the fault-absorption mechanism operates consistently: RMSE decreases with team size and remains bounded across all trajectory types. Hover RMSE is 19% lower than figure-eight (39.75 vs. 48.84 cm for N=3), confirming that the figure-eight amplifies post-fault error through cable-pendulum near-resonance.

E. Cable Discretization Sensitivity

The bead-chain cable model discretizes each continuous cable into n_b point-mass beads connected by $(n_b{+}1)$ tension-only spring-damper segments. All preceding results use n_b=8. A dedicated sweep over $n_b \in \{2, 4, 8, 12, 16\}$ in the N=5 cascading-fault scenario kept cable-level stiffness and damping invariant while varying spatial resolution. Across the full range, the first-fault peak error changed by only 0.3 cm (31.5–31.8 cm), the second-fault peak by 0.2 cm (43.2–43.4 cm), the whole-run RMSE by 0.1 cm (43.1–43.2 cm), and the recovery times remained 4.09 s and 3.67 s for the two faults. The main fault-tolerant transport conclusion is therefore insensitive to cable discretization over the tested range, and the n_b=8 baseline remains the practical choice.

VII. Discussion and Conclusion

This section interprets the results, identifies the mechanism and its limits, situates the findings relative to prior work, and outlines what remains open.

The central finding and its mechanism: The full-Drake results (Tables I–III) show that the tested GPAC path maintains transport through one to three abrupt cable-snap faults without explicit fault detection or reconfiguration, but with substantial persistent degradation. In the canonical scenarios, post-fault RMSE is 44–49 cm; in matched single-fault tests it falls from 48.8 to 38.4 cm as N increases from 3 to 7. These error levels are about three payload radii and roughly 40% of a nominal cable length, so the contribution is transport continuity rather than precision placement. Relative to the Tether_Lift baseline [13], this is Tether_Grace's central novelty: the same N-agnostic local-sensing architecture that enabled decentralized cooperative lift also remains well-defined under abrupt topology change. Proposition 1 explains this only for the simplified per-agent model, and the tension feedforward ablation (Table VI) pinpoints the mechanism in the reported runs: PID feedback combined with gravity feedforward, not cable-tension feedforward ($<1\%$ RMSE contribution). Gravity feedforward provides the baseline hover thrust requiring no knowledge of other agents, PID feedback corrects the tracking-error displacement, and the anti-windup-limited integral term carries the increased load share. Section IV predicts a persistent offset of about 0.37 m from that clamp, consistent with the reported terminal errors. That cable-tension feedforward contributes so little was itself counter-intuitive; the authors initially hypothesized it as the primary mechanism.

Why the result is non-trivial: While PID with integral action rejects bounded step disturbances in single-input linear systems, the extension to multi-agent systems with elastic cable coupling, underactuation, and unobservable topology changes is not obvious. The gap between the scalar-model prediction ($\sim$18 cm) and full-plant observation ($\sim$107 cm)—a factor $\alpha \approx 5.8$—quantifies the aggregate contribution of effects the scalar model omits: elastic cable coupling (the bead-chain produces force oscillations the scalar model treats as smooth), attitude-loop delay ($\sim$37% bandwidth overlap between the 5 ms inner-loop period and the $\sim$8 ms force-transmission timescale), and multi-axis interaction through R_i. The small-gain analysis confirms that a proper multivariable treatment is needed to close this gap formally, while the simulation evidence confirms qualitative consistency with the scalar prediction. Additional secondary effects operate within the fault-tolerant envelope without altering the primary mechanism:

- *Anti-swing damping* ($k_{\mathrm{sw}} = 0.5$) is small compared to PID forces during the fast cable-snap impulse but attenuates the slower (1–2 s) post-fault pendular oscillation between trajectory direction changes.
- *Cable stiffness* k_s affects transient duration and wave speed ($v_{\mathrm{wave}} \propto \sqrt{k_s}$) but not impulse magnitude, which is set by $T_j \approx m_L g/N$. Higher stiffness produces a sharper but not larger transient; the finite-time bound (29) accommodates this variation through τ_{fault}.
- *Freed-agent escape* was revised during evaluation after the original parameters required $\sim 60^\circ$ tilt (below-gravity vertical thrust). The revised two-waypoint design is conservative but reliable; a systematic parametric study falls outside this paper's scope.

Design tools and their validation: The $f_{\max}$ boundary sweep (Table VII) validates the thrust-margin screen by quantifying the exact failure boundary at $\sim$45 N, coinciding with the theoretical minimum (34) with the default $f_{\max} = 150$ N providing $3\times$ headroom. This yields an actionable design criterion: $f_{\max} > (Nm_Q + m_L)g/(N-k) + f_{\mathrm{track}}$. The matched single-fault data (Table III) further shows that peak error is nearly constant across team sizes ($\sim$107–112 cm) while RMSE improves because larger teams recover faster during mid-trajectory tracking. Together with the closed-form cost ratio $\rho_{\mathrm{FT}} = N/(N-k)$ and the qualitative transient prediction, these results provide practical screening tools for actuator sizing and expected degradation in the tested architecture rather than a closed, fully general design methodology.

From hypothesis to structural property: Proposition 1 identifies topology-independent error dynamics for a simplified model, and four lines of empirical evidence support qualitative extension to the full plant: thrust margins (1.75–2.33 capacity ratios), monotonically decreasing RMSE with N under matched single-fault comparison, a quantified thrust boundary at the theoretical minimum, and fault-count isolation at N=7. Three formal gaps remain (Section IV-A): impulsive slack-to-taut forces, attitude-loop delay under fast transients, and the cable-force small-gain condition. Closing these—particularly the $\mathcal{H}_\infty$-based small-gain verification via Drake's `Linearize()`—remains the primary theoretical open problem. The current paper therefore supports a simulation-validated architectural explanation, not a closed nonlinear stability theorem.

Comparison with prior work: Liang et al. [10] address suspension failure via centralized reconfiguration with known failure identity; the present work shows neither assumption is necessary to maintain transport, achieving stable tracking (44–49 cm RMSE) without failure identification or communication. Zeng et al. [11] observed implicit resilience in MARL-based cable transport but did not explain the mechanism. Proposition 1 supplies a partial structural explanation for the tested PID-plus-gravity-feedforward architecture and suggests that closely related N-agnostic local-feedback architectures may inherit similar behavior, but that broader generality is not established here. The centralized oracle (Table VIII) achieves comparable performance because the tested oracle adds only perfect post-fault load-share feedforward; it should be interpreted as a weak information baseline, not as a globally reconfigured centralized optimum.

Remaining open problems: Five directions remain for future work:

- *Adaptive residual integration.* An $\mathcal{L}_1$ adaptive controller [19], [20] is implemented in the codebase. Its key property for topology invariance is that $\hat{d}$ carries no pre-fault history (unlike CL's buffer of M=50 pairs). Head-to-head comparison with PID-only under ESKF+wind conditions is planned as near-term future work.
- *Integral gain sensitivity.* The integral time constant $1/k_i \approx 6.7$ s sets the post-fault recovery timescale and determines ρ_{FT}. A sweep of k_i would characterize the tradeoff between faster recovery and transient overshoot, complementing the $f_{\max}$ boundary sweep.

- *Formal extension to the full plant.* Three self-contained technical problems remain: bounding impulsive slack-to-taut forces, analyzing cascaded position/attitude stability under fast transients, and verifying the small-gain condition for cable-force feedback.
- *Cable pendulum management.* The figure-eight trajectory excites near-resonance ($\sim$2 s cable pendulum period vs. $\sim$3 s corner interval). Trajectory shaping with pendulum frequency avoidance may be needed for sustained aggressive maneuvering.
- *Hardware validation.* All results are simulation-based. Physical multi-quadrotor cable transport with actual cable-snap events remains the ultimate test.

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